JEFFREY OCK

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Bioprocess engineer working in GMP gene therapy CMC development, with a research background in tissue engineering, bioreactor design, and ML-based image analysis. Continuing ML hobby projects in free time.

PROFESSIONAL EXPERIENCE

Krystal Biotech, Inc. **Jul 2026 - Present**
Bioprocess Engineer I *Pittsburgh, PA*

- CMC development through GMP upstream and downstream process development for novel gene therapy products spanning respiratory, ophthalmology, dermatology, and oncology programs

RESEARCH EXPERIENCE

Stem Cell & Tissue Engineering Lab, Texas A&M University **May 2024 - May 2026**
Research Technician | PI: Feng Zhao, PhD *College Station, TX*

- Designed and built a dermal matrix bioreactor for scaled-up cell culture of ECM samples; delivered reproducible output across 10 consecutive 5 x 5 cm cell sheets with no significant batch-to-batch variability
- Led development of an expandable acellular dermal matrix scaffold seeded with primary human dermal fibroblasts (hDFs), achieving 67% greater area expansion versus control
- Led a micropatterned-substrate project for an anisotropic interwoven skin graft; demonstrated >200% increase in tensile strength along with elevated senescence and proto-myofibroblast differentiation
- Characterized primary and hiPSC-derived cardiac fibroblasts on anisotropic scaffolds and ran cell migration studies across scaffold porosities
- Trained 2 undergraduate and 1 graduate researcher in cell culture, confocal microscopy, tensiometry, plasma cleaning, and EtO sterilization; managed lab inventory and equipment

Forster Lab, UPMC Children's Hospital of Pittsburgh **May 2023 - Aug 2023**
Student Research Internship Program (SRIP) | PI: Catherine Forster, MD *Pittsburgh, PA*

- Showed morphological and proteomic congruence between Chrm3-/- and spinal cord injury (SCI) mouse models of neurogenic bladder; presented at the SRIP Poster Symposium
- Processed tissue through paraffin embedding, microtome sectioning, IHC co-staining, and H&E staining to characterize urothelial phenotypes, with complete lab documentation
- Performed IACUC-compliant mouse work: colony maintenance, tissue harvest, and euthanasia of Chrm3 knockout and wild-type lines

COMPUTATIONAL & ENGINEERING PROJECTS

CNN Cell Counting Model for Fluorescence Microscopy **Jul 2023**
Software Lead | UPMC Children's Hospital of Pittsburgh *Pittsburgh, PA*

- Designed and trained a convolutional neural network in PyTorch to count cells in cytoplasmic-stained fluorescence images; reached MAE of 79.8 on n = 58 images (~400 cells/image)

CytoCaricature: Cell Image Analysis Application **Aug 2024 - May 2026**
Software Lead | Texas A&M University *College Station, TX*

- Built a Python pipeline for morphometric cell analysis: channel isolation, Gaussian filtering, intensity thresholding, watershed segmentation, and nuclear shape index (NSI) heatmaps

VIPER Bioreactor (Versatile Innovative Peristalsis Environment bioReactor) **Aug 2024 - May 2026**
Electrical & Software Lead | Texas A&M University *College Station, TX*

- Co-designed a peristaltic bioreactor reproducing the flow strain and axial stress of peristaltic organs
- Designed electrical safety circuits and wrote a Python CLI for real-time control of flow rate and peristalsis parameters

PUBLICATIONS & PRESENTATIONS

- Sharma, A., Sharma, D., Horn, S.J., Wen, Z., **Ock, J.**, et al. (2026). Human cell-derived dermal-specific interwoven extracellular matrix for diabetic wound healing. *Acta Biomaterialia*.
- **Ock, J.**, Proca, A., Stout, T., Wang, E., Liu, S., Zhao, F. Surface treatment of PDMS substrates for the development of an expandable dermal extracellular matrix. In preparation.
- **Ock, J.**, Stout, T., Wang, E., Kaluarachchi, S., Wen, Z., Liu, S., Yang, Y., Zhao, F. Enhancing the mechanical strength and altering properties of dermal ECM through micropattern design. In preparation.
- Poster: **Ock, J.** et al. "ECM morphology and cellular mechanobiology on interwoven vs. interwoven-aligned PDMS scaffolds." LAUNCH Undergraduate Research, Texas A&M, Jul 2025.

- Poster: **Ock, J.** et al. "Knockout muscarinic acetylcholine receptor subtype M3 in mice as a model for urothelial injury due to neurogenic bladder." SRIP Symposium, Pittsburgh, Jul 2023.

EDUCATION

Texas A&M University, College of Engineering **May 2026**
B.S. Biomedical Engineering | GPA: 3.48/4.00 *College Station, TX*

Relevant Coursework: Biomedical & Health Data Science, Biostatistics, Biomolecular Engineering, Drug Delivery, Tissue Engineering, Nanotechnology, Molecular & Cell Biomechanics, Imaging Living Systems, Biomaterials

Teaching Assistant (Aug 2025 - May 2026): BMEN 351, Machine Learning & Data Science in BME (office hours and grading for 100+ students on ML model development and data analysis); BMEN 344, Biological Interactions: Medical Devices & Host (100+ students)

Honors: Kidney STARS Award, American Society of Nephrology (2023); National Merit Finalist (2022); President's Endowed Scholarship (2022)

SKILLS

Bioprocess & Quality: GMP, GDP, unidirectional cleanroom use/maintenance, commercial fixed-bed bioreactors, drug product filling with isolator, autoclave, commercial large scale mammalian cell culture, commercial filtration methods

Cell & Tissue Culture: Mammalian and primary cell culture (BSL-2), hDF and hiPSC-derived cells, 3D culture, decellularization, ECM scaffold fabrication, PDMS fabrication, silicon wafer nanofabrication (RIE lithography), EtO sterilization, plasma treatment

Assays & Imaging: IHC, IF, H&E, DNA, MTT, and Live/Dead assays; confocal (Olympus FV1000) and fluorescence microscopy (Zeiss, Leica); high-content imaging; paraffin embedding and microtomy; contact angle tensiometry

ML & Data: Python (PyTorch, scikit-learn, NumPy), R, MATLAB, SQL/PostgreSQL; CNNs, image segmentation, statistical analysis, data visualization; GraphPad Prism, Minitab

Software & Hardware: JavaScript/TypeScript, Node.js, C/C++, Java; SolidWorks, Autodesk Inventor, LabVIEW, PCB design (EasyEDA), Arduino, circuit assembly, 3D printing

Certifications: CITI (active through May 2027): Responsible Conduct of Research; Conflict of Interest (PHS); Working with Mice in Research; Working with Small Animals; Controlled Substances in Animal Research

Languages: English (native), Korean (native)